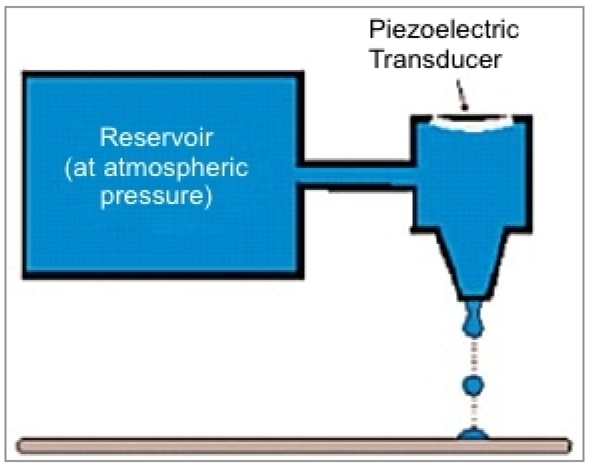

Fig. 2: Arrangement of ink-jet printing

on type of greenhouse, type of crop, environmental control facilities and so on.

- Reliability of crop increases under greenhouse cultivation.
- Ideally suited for vegetables and flower crops.
- Year-round production.
- Off-season production of vegetables, flowers and fruits.
- Disease-free and genetically-superior transplants can be produced continuously.
- Efficient utilisation of chemicals, pesticides, etc to control pests and diseases.
- Water requirements of crops is limited and easy to control.
- Use of modern techniques like hydroponics, aeroponics and nutrient film techniques are possible only under greenhouse farming.

Cost of a greenhouse with printed TOPV roof

A printed TOPV is quite affordable now. It costs about ₹ 465 per square metre; whereas, cost of a UV-resistant 200-micron polythene sheet is about ₹ 60 per square metre. In India, fixed cost of a greenhouse having poly-sheet roof using medium technology with fan system, cooling pads and arrangement for misting to maintain humidity is about ₹ 1100 per square metre, including land and development. If the same is installed with printed TOPV roof, cost would be about ₹ 1505 per square metre. In both cases, recurring cost is about ₹ 1660 per square metre.

Efficiency of TOPVs is around ten per cent. Considering the safety factor, we can assume conversion efficiency to be six per cent and insolation received at Earth's surface to be 1000W per square metre.

With these assumptions, TOPVs can produce 60W per square metre of electricity at a cost of about ₹ 6 per kWh, which is highly competitive as compared to the present cost of power generation by burning fossil fuels. Hence, marriage between these two technologies would yield many additional benefits including generation of electricity as a by-product. Moreover, the government may likely give a subsidy of fifty per cent of the total project cost. Payback period is about three years.

Organic solar cell

An organic solar cell or organic photovoltaic (OPV) is a type of photovoltaic that uses conductive organic polymers for light absorption to produce electricity from sunlight by photovoltaic effect. Molecules used in OPV are solution-processable at high throughput and are cheap, resulting in low production costs to fabricate in large volumes.

Combined with the flexibility of organic molecules, organic solar cells are potentially cost-effective for photovoltaic applications. Molecular engineering, that is, changing the length and functional group of polymers, can change the band gap, allowing for electronic tunability.

Optical absorption coefficient of organic molecules is high, so a large amount of light can be absorbed with a small amount of active materials with a film thickness usually in the order of hundreds of nanometres.

The main disadvantage is low efficiency. As of 2015, OPV was supposed to achieve over ten per cent efficiency. In 2018, a record-breaking efficiency of 17.3 per cent was achieved via tandem structure.

OPV is lightweight, transparent and can be made flexible. Ink-jet printing is used for deposition of the active layer of bulk heterojunction (BHJ) solar cells, which are superior. This reduces the cost of production. Moreover, the ink-jet process yields films that are pin-hole-free and therefore suitable as the active layer in organic solar cells.

Active materials used in OPV. Typically, active layer of an organic photovoltaic cell is composed of a blended film of conjugated polymer as electron donor and fullerene derivative as acceptor. Some of the blends are: poly (3-hexylthiophene)—in short, P3HT—mixed with fullerene derivative like phenyl-C61-butyric acid methyl ester (PCBM); and p-DTS (FBTTh2) 2 mixed with fullerene derivative PC71BM, etc.

C60N as a dipole additive may be mixed, which increases power conversion efficiency due to more efficient extraction of photogenerated charge carriers. An ink made from a mix of polymer and fullerene derivative is used for ink-jet printing. Ink is then printed on a flexible substrate in 0.2-micrometre thickness to absorb Sun's energy (Fig. 2). Using this incredible thin layering, one kilogram of polymer can be used to print a solar cell, the size of a football field.

Physics behind organic solar cells

A typical OPV device consists of a photoactive layer sandwiched between two electrodes, one of which has to be transparent to allow incoming light to reach the photoactive layer. Absorption of light excites the electron into the lowest-unoccupied molecular orbital (LUMO) of absorber material, creating strongly-bound electron-hole pairs called excitons.

Excitons must overcome their binding energy to dissociate. In the absence of a mechanism to dissociate excitons, these will spontaneously decay. In an organic bilayer device employing, say, copper phthalocyanine (CuPc)/perylene derivative, difference in electronic affinities between these two materials creates an energy offset at their interface, driving exciton dissociation.

Similar to this early device, many OPVs now have an active layer consisting of two materials, one electron donor that is a conjugated material and the other electron acceptor, which is a fullerene derivative.

Conjugated polymers are a particularly interesting class of organic semiconductors, as polymer conductivity can be easily and reversibly varied by doping, resulting in a range in conductivity from insulating to metallic. PEDOT, acronym of poly (3, 4-ethylenedioxythiophene), P3HT: poly (3-hexylthiophene), PANI (polyaniline), etc are conjugated polymers.

A fullerene is an allotrope of carbon whose molecules consist of